

# North Star Second Brain

Assignment 2 submission | Evidence captured 22 August 2026

**A deterministic personal information pipeline that collects real job, learning, RSS, and GitHub sources, then ranks them against one personally chosen career goal.**

**Applicant goal:** move from Data Engineer to Generative AI Engineer by April 2027, using evidence from real work, learning, and opportunities.

# 1. North Star statement

By April 2027, starting from my current role as Data Engineer, I will be working as a Generative AI Engineer at a company, verified by an employment start date and an official offer letter, so I can make production AI engineering my next career focus.

Four parts a stranger can check:

| Part              | What it says                                                                                                      |
|-------------------|-------------------------------------------------------------------------------------------------------------------|
| Current state     | I am currently a Data Engineer.                                                                                   |
| Time-bound goal   | The deadline is April 2027.                                                                                       |
| Provable activity | I will be employed as a Generative AI Engineer, checked using an employment start date and official offer letter. |
| Reason            | I want production AI engineering to become my next career focus.                                                  |

## 2. Source list

The source list is based on sources actually connected to this project. The two strongest signals are marked with [MOST SIGNAL].

| Source                                     | What enters the brain                                                                                                                                       | Signal judgment                                                  |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| LinkedIn Job Alerts via IMAP [MOST SIGNAL] | Real job-alert emails, including multiple jobs per digest.                                                                                                  | Primary opportunity signal.                                      |
| Zapier structured job email [MOST SIGNAL]  | Structured title, company, location, URL, date, tags, and responsibilities.                                                                                 | Primary clean job signal.                                        |
| GitHub repository search                   | GenAI repositories searched by topics and sorted by stars or updates.                                                                                       | Strong practical skill and trend signal.                         |
| RSS feed                                   | Public GenAI engineering articles and posts.                                                                                                                | Useful knowledge signal; not employment proof.                   |
| LMS folder                                 | Course notes and exported learning material.                                                                                                                | Personal learning signal.                                        |
| Consciously cut                            | Personalized LinkedIn home feed: no stable official RSS route and mixed signal. Google security and Pipedream notices: operational mail, not career inputs. | Deliberately excluded; job alerts and public GenAI feeds remain. |

## 3. Second brain folder and real items

The project stores raw Markdown separately from normalized and synthesized outputs. Current evidence counts:

| Folder                   | Items | Purpose                               |
|--------------------------|-------|---------------------------------------|
| second_brain/raw/jobs/   | 33    | Raw email and RSS job material.       |
| second_brain/raw/github/ | 81    | Raw GitHub repository search results. |

| Folder                               | Items | Purpose                        |
|--------------------------------------|-------|--------------------------------|
| second_brain/raw/learning/           | 4     | Raw LMS Markdown notes.        |
| second_brain/normalized/jobs/        | 53    | Structured job records.        |
| second_brain/normalized/github/      | 81    | Structured repository records. |
| second_brain/normalized/learning/    | 3     | Structured learning records.   |
| second_brain/normalized/jobs_scores/ | 65    | LLM ranking output.            |

## Folder tree:

```
second_brain/
  raw/
    jobs/
    github/
    learning/
  normalized/
    jobs/
    github/
    learning/
    jobs_scores/
```

Evidence capture: generated from the current repository folders.

Repository evidence | second\_brain/

```
second_brain/
  raw/jobs/          33 Markdown files
  raw/github/        81 Markdown files
  raw/learning/      4 Markdown files
  normalized/jobs/    53 JSON files
  normalized/github/  81 JSON files
  normalized/learning/ 3 JSON files
  normalized/jobs_scores/ 65 JSON files

Generated from the current workspace on 22 August 2026.
```

**Examples of real ingested items:** LinkedIn Generative AI Engineer alerts, Senior Data Engineer from Zapier, Scaling LLM-based Ranking Systems with SGLang, Microsoft Generative AI for Beginners, LangChain, RAGFlow, vLLM, LMS RAG practical codepath, LMS LLM memory notes, and LMS Connecting the Dots notes.

## 4. Personalization engine

The engine uses one scoring prompt in **deterministic\_ingestion/SCORE\_PROMPT.md**. It asks GROQ to score every normalized item against the North Star using five weighted dimensions:

| Dimension               | Weight | Meaning                                                              |
|-------------------------|--------|----------------------------------------------------------------------|
| Alignment               | 40     | Direct connection to becoming a Generative AI Engineer.              |
| Impact on goal          | 20     | Likelihood of moving toward employment or strong portfolio evidence. |
| Feasibility / proximity | 15     | Achievability from the current Data Engineer role.                   |
| Skill match             | 15     | Presence of LLM, RAG, inference, MLOps, and related skills.          |
| Urgency / timing        | 10     | Timeliness and opportunity window.                                   |

The LLM returns JSON with **score**, **breakdown**, **confidence**, and short **notes**. GitHub repositories are judged for practical implementation and engineering depth; LMS items for hands-on learning; jobs for role fit; RSS for learning and trend signals.

### Before and after example

**Before: raw incoming Zapier email**

```
--- source: email date: 2026-08-21T13:01:44+00:00 raw_id: job-senior-data-engineer-acme-corp-remote-no-reply-sbhc9p-zapierm
```

**After: normalized record**

```
{
  "source": "email",
  "email_type": "zapier_single_job",
  "title": "Senior Data Engineer",
  "company": "Acme Corp",
  "location": "Remote",
  "url": "https://www.linkedin.com/jobs/view/123456",
  "date_posted": "2026-08-20",
  "tags": [
    "data-engineer",
    "spark",
    "python"
  ],
  "skills": [
    "etl",
    "python",
    "spark"
  ],
  "responsibilities": [
    "design etl",
    "manage streaming",
    "mentor",
    "design ETL, manage streaming, mentor junior engineers. Apply: https://www.linkedin.com/jobs/view/123456"
  ]
}
```

**Filtered/scored output example**

```
{
  "score": 73,
  "breakdown": {
    "alignment": 65,
    "impact": 70,
    "feasibility": 80,
    "skill_match": 90,
    "urgency": 70
  },
  "confidence": 0.9,
  "notes": "Open-source repo with strong LLM and agent focus, useful for portfolio; high feasibility and skill match."
}
```

**Code proof: the scoring stage scans all three normalized source folders:**

```
NORM_DIRS = [NORM_ROOT / name for name in ('jobs', 'github', 'learning')]
files = sorted(path for norm_dir in NORM_DIRS
                for path in glob(str(norm_dir / '*.json')))
output = call_llm(build_prompt(north_star, item))
parse_and_save(output, item_id)
```

Evidence capture: actual pipeline fields used in this submission.

### ●●● Evidence capture | raw -> normalized -> scored

```
RAW EMAIL
title: "Senior Data Engineer"
company: "Acme Corp"
url: https://www.linkedin.com/jobs/view/123456

NORMALIZED JSON
title: Senior Data Engineer | company: Acme Corp
skills: [etl, python, spark]

SCORED JSON
score: 37 | confidence: 0.5
breakdown: alignment, impact, feasibility, skill_match, urgency
```

## 5. Ingestion mechanism

The pipeline is executable both manually and through GitHub Actions. The weekly workflow runs every Monday at 06:00 UTC and can also be started manually with `workflow_dispatch`.

| Step                | Implementation evidence                                                                                  |
|---------------------|----------------------------------------------------------------------------------------------------------|
| GitHub repositories | <code>ingest_github.py</code> searches GenAI repositories and writes raw Markdown.                       |
| RSS knowledge       | <code>ingest_linkedin_rss.py</code> reads configured RSS/Atom feeds and deduplicates entries.            |
| Email jobs          | <code>ingest_email_imap.py</code> fetches unseen messages and filters unrelated security/Pipedream mail. |
| LMS                 | <code>ingest_lms_folder.py</code> imports files from <code>inbox_lms/</code> .                           |
| Normalization       | <code>normalize_job.py</code> handles jobs; <code>normalize_sources.py</code> handles GitHub and LMS.    |
| Scoring             | <code>score_with_llm.py</code> scans all normalized folders and calls GROQ.                              |

**Workflow evidence:** `.github/workflows/weekly-ingest.yml`. **Manual schedule evidence:** `deterministic_ingestion/cron_example.txt`.

```
0 * * * * ingest_linkedin_rss.py
15 * * * * ingest_github.py
30 * * * * ingest_email.py
```

### Code proof: the workflow runs the collectors before normalization and scoring:

```
python deterministic_ingestion/ingest_email_imap.py
python deterministic_ingestion/ingest_linkedin_rss.py
python deterministic_ingestion/ingest_github.py
python deterministic_ingestion/ingest_lms_folder.py
python deterministic_ingestion/normalize_job.py
python deterministic_ingestion/normalize_sources.py
python deterministic_ingestion/score_with_llm.py
```

Evidence capture: workflow and manual command sequence.

#### Evidence capture | ingestion mechanism

```
GitHub Actions: .github/workflows/weekly-ingest.yml
schedule: 0 6 * * 1 (Monday 06:00 UTC)

python deterministic_ingestion/ingest_email_imap.py
python deterministic_ingestion/ingest_linkedin_rss.py
python deterministic_ingestion/ingest_github.py
python deterministic_ingestion/ingest_lms_folder.py
python deterministic_ingestion/normalize_job.py
python deterministic_ingestion/normalize_sources.py
python deterministic_ingestion/score_with_llm.py
```

The raw records remain Markdown. The normalizers create structured JSON. Only the final scoring step calls GROQ, and its output is stored separately in jobs\_scores.

End of submission evidence.